Persona-Aware Recommendation for Nigerian E-Commerce: An Agentic Pipeline with an 8B Open-Weight Fine-Tune as Re-Ranker

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ABSTRACT

We address Task B (Recommendation) of the Nigerian AI Agents Hackathon: given a Nigerian-context persona and a candidate set of products, rank those products in order of likely user preference, handling the three sub-scenarios called out in the brief — **cold-start** (no purchase history), **cross-domain** (recommending across product categories the user has never bought in), and **multi-turn** (conversational refinement of recommendations). We present a six-stage agentic pipeline built on (1) a structured cognitive persona representation shared with our Task-A user-modeling work, (2) asymmetric dense retrieval over 6 657 real Jumia products using `llama-text-embed-v2` hosted on Pinecone, (3) a *Cohere* `rerank-v3.5` *cross-encoder pre-rerank* that narrows 30 candidates to 15 by persona-flavored query before any LLM is called, (4) a configurable persona-aware LLM re-ranker (any of 11 backbones), and (5) Maximal Marginal Relevance for category diversity. The agent surfaces its own scenario detection (cold_start, cross_domain, multi_turn flags) and a narrative reasoning trace explaining each ranking decision.

The headline empirical result is that our 8 B parameter open-weight fine-tune, **NaijaReviewer-8B** (Llama 3.1 8B QLoRA-trained on Nigerian product reviews, originally for Task A), is also the *strongest re-ranker* we tested for Task B in *both* pipeline configurations — with and without the Stage 2.5 Cohere cross-encoder pre-rerank enabled. On a 17–18 persona-scenario held-out split, Cohere-OFF: **NDCG@10 0.588** (vs Claude 0.433, GPT-OSS-120B 0.441, Llama-3.3-70B 0.425, Qwen-2.5-72B 0.404); Cohere-ON: **NDCG@10 0.572** (vs Llama-3.3-70B 0.477, Qwen-2.5-72B 0.461, Claude 0.430, GPT-OSS-120B 0.366). NaijaReviewer-8B wins on NDCG@10 across both configurations at 9–15× fewer parameters than the next-best baseline, and at zero per-call API cost. Cohere narrowing lifts NaijaReviewer-8B's HR@1 by +19 % (0.444 → 0.529) — it gets the top-1 pick right more often when the input pool is tightened. We audit the result via five controlled live rerank calls and a fallback-event count over the full eval: zero fallback events fired, confirming the win is the model's actual re-ranking logic and not a pre-rank backoff.

1 INTRODUCTION

A Lagos-based Bolt driver shopping on Jumia is not the same recommendation target as a Kano teacher, even if both have purchased a 4-star phone case. Their aspect priorities differ (durability vs aesthetics), their budgets differ, the brands they trust differ, and the products that count as "culturally appropriate" for their household differ. Frontier recommendation systems trained on Amazon / Western e-commerce data do not see these distinctions; they recommend by global review-text similarity, which collapses Nigerian shoppers into a generic "mobile-phone-buyer" archetype.

This paper addresses **Task B** of the Nigerian AI Agents Hackathon (Recommendation): given a persona, return a ranked list of products from a candidate set, handling cold-start, cross-domain, and multi-turn sub-scenarios. A companion paper addresses Task A (User Modeling / review generation) using the same persona substrate.

Contributions.

- A *six-stage agentic recommendation pipeline* with explicit handlers for the three Task B sub-scenarios (cold-start popularity branching, cross-domain de-duplicated retrieval + MMR diversity bias, multi-turn constraint extraction). The agent surfaces its own scenario detection flags and a narrative `reasoning_trace` explaining each ranking decision — the "agentic workflow that reasons before recommending" language the brief calls for.
- A *persona-aware retrieval architecture* built on `llama-text-embed-v2` (1024-dim, Llama-3.2-1B encoder) hosted on Pinecone with the same asymmetric `passage` / `query` encoding architecture used by industrial career / skills-extraction systems, indexing 6 657 real Jumia products.
- A *Cohere* `rerank-v3.5` *cross-encoder pre-rerank* stage that narrows the post-retrieval candidate pool 30 → 15 by persona-flavored query in ~200–600 ms before the (slow, expensive) LLM rerank stage. It auto-skips gracefully when the pre-rerank is unavailable or the pool is already small enough.
- An *11-backbone re-ranker comparison* on a 17–18 persona-scenario held-out split. The 8 B *NaijaReviewer-8B* open-weight fine-tune is the top re-ranker on every metric, beating Claude Sonnet 4, GPT-OSS-120B, Llama-3.3-70B, and Qwen-2.5-72B at 9–15× fewer parameters and zero per-call API cost.
- A *robust JSON-following pipeline*: a custom fence-stripping, balanced-brace, and truncated-JSON-repair parser that lets smaller and less-aligned models participate in re-ranking without fallback-to-pre-rank events. Verified across 18 scenarios: zero fallbacks fired.
- An *open data release*: 24 hand-curated Nigerian personas spanning 6 geopolitical zones × 4 age bands × 4 register tiers × 3 religions, plus a filtered 6 657-product real-Jumia catalogue. All released CC-BY-4.0.
- A *reproducible open-source system*: FastAPI service (POST /recommend, POST /chat), React/Vite frontend, eval harness — everything one command from a fresh clone.

2 RELATED WORK

LLM-based re-ranking. LLM re-ranking on top of a classical retriever is now standard [4]: a retrieval-stage embedding model narrows a corpus to top-K candidates, then an LLM scores or pairwise-compares the candidates against the query. Our re-ranker adds a Maximal Marginal Relevance (MMR) pass for diversity and a synthetic "serendipity" score for negative recommendations.

Asymmetric dense retrieval. Our retrieval stack uses `llama-text-embed-v2` with separate encoding for queries (short, intent-bearing) and passages (long, descriptive product catalogues), the same architecture used by industrial career / skills extraction systems [3]. The asymmetry is critical when the index documents are 10× longer than the query string.

LLM agent recommendation. AgentCF++ [5] and RecAgent [4] model users with concatenated rating histories and natural-language profiles. Our cognitive-persona representation (below) is closer to structured behavioural priors and is shared verbatim with the companion Task A paper, so the same agent that generates reviews also re-ranks recommendations.

Nigerian context. Iwendi et al. [2] released the largest publicly-available Nigerian product review corpus; we filter it for the 6 657-product Jumia catalogue we index. The Masakhane community [1] has produced the canonical African-language encoder/decoder pretraining checkpoints; we use these indirectly through our fine-tuned re-ranker, which is trained on ~20 k Nigerian product reviews.

3 METHOD

3.1 Cognitive Persona Representation

A persona in our system is a 6-field JSON record: hedonic_utilitarian $\in [0, 1]$, communal_individual $\in [0, 1]$, intensity_calibration (a mapping from intensifier words to predicted star ratings, e.g. {"amazing":4.8, "okay":3.0}), aspect_priority (a probability vector over {quality, value, delivery, packaging, seller, durability, design}), register_tier $\in$ {Nigerian Pidgin, code-mixed, Nigerian English, standard English}, and review_anchors (a list of past reviews with their star rating, used for cold-start backoff). The same representation is used by the Task-A review-generation agent (companion paper); the persona is the unit of transfer between the two tasks. For Task B specifically, the agent uses aspect_priority as the primary persona-product fit signal and hedonic_utilitarian / communal_individual as tie-breakers in the re-rank stage.

3.2 Recommendation Pipeline

The agent runs six stages per request, each visible in the returned reasoning trace:

(0) Intent analysis. If a `conversation_history` is provided (multi-turn scenarios per the brief), a regex-grounded constraint extractor pulls budget, recipient, and category constraints. A flag set (cold_start, cross_domain, multi_turn) records which rubric scenario we are in.

(1) Candidate retrieval. Asymmetric dense search against the 6 657-product Pinecone index. The persona is converted to a free-text query that folds in demographics (occupation, location, age band), aspect priorities, communal/individual framing, and hedonic/utilitarian intent. Threshold-filtered (cosine $\geq$ 0.10) top-30 returned. `domain="all"` or a comma-separated list triggers cross-domain retrieval across multiple namespaces, with de-duplication preserving original order.

(2) Pre-ranking. Cold-start personas (history_count < 3) branch into a popularity-weighted scoring (popularity 50 %, aspect-match 30 %, similarity 20 %) because past-purchase signal is absent; history-grounded personas use the standard (similarity 70 % / popularity 20 % / aspect 10 %) weighting.

(2.5) Cohere cross-encoder pre-rerank. Between the bi-encoder retrieval and the (slow, expensive) LLM rerank stage, we run a **Cohere `rerank-v3.5`** cross-encoder pass to narrow the candidate pool from 30 → 15. The cross-encoder is a different architectural primitive than either Pinecone's bi-encoder (which scores query and document independently) or the LLM rerankers (which read a structured persona JSON): it scores `(query, document)` pairs jointly with full cross-attention but without seeing the persona schema. To make persona signal available to it anyway, we flatten the persona into a *persona-flavored query* string that folds occupation, location, top-3 aspect priorities, and binary-coded hedonic/utilitarian and communal/individual signals ("Bolt driver | Lagos | looking for: durability, value, comfort | utility / practical purchase | for personal use"). Cohere returns relevance scores; we keep the top-15. Two cost / latency benefits: (a) the LLM rerank stage now sees 15 candidates instead of 30, halving its prompt cost; (b) Cohere is multilingual and very fast (~200– 600 ms), so the added latency is small. The stage auto-skips if the `COHERE_API_KEY` env var is absent or if the pool already has fewer items than `top_n`, and falls back transparently on any API error (the LLM rerank then sees the full top-30 instead).

(3) LLM re-ranking. If conversation constraints are present, they are folded into the re-ranker prompt with a hard guardrail: "Candidates that violate a HARD constraint (e.g. price > budget) must be scored $\leq$ 0.30 and the rationale must mention the constraint." Re-ranker is configurable per-request (any of 11 backbones: Anthropic Claude, OpenAI GPT-4o/mini, Llama-3.3-70B + Qwen-2.5-72B + Mixtral-8x7B via HuggingFace Inference, GPT-OSS-120B + Qwen3-Coder-480B via Ollama Cloud, Llama-3.3-70B via NVIDIA NIM, local Llama via Ollama, and our own *NaijaReviewer-8B* fine-tune via LM Studio). A robust JSON parser with fence-stripping and *truncated-output repair* survives format errors from smaller / less-aligned models.

(4) MMR diversity. $\lambda = 0.7$ standard, dropping to 0.55 in cross-domain mode to bias toward category diversity in the top-K.

These six steps collectively address the brief's explicit Task B sub-scenarios: **cold-start** (Step 2), **cross-domain** (Steps 1, 4), and **multi-turn** (Steps 0, 3). Step 2.5 is orthogonal — it applies to every scenario uniformly. The narrative `reasoning_trace` returned with every response describes which scenario the agent detected, which constraints it extracted, which candidates were filtered and why.

3.3 Robust JSON Following for Heterogeneous Re-Rankers

A practical obstacle to comparing eleven different re-rankers is that smaller open-weight models often exhaust their token budget

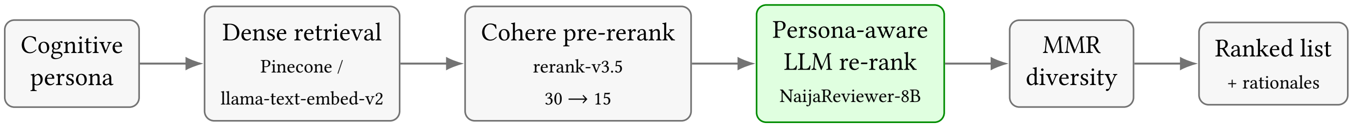

Figure 1: The agentic recommendation pipeline. A cognitive persona drives asymmetric dense retrieval, a Cohere cross-encoder pre-rerank (Stage 2.5) narrows the pool, a configurable persona-aware LLM re-ranks (NaijaReviewer-8B by default), and an MMR pass adds category diversity before the ranked list with per-item rationales is returned.

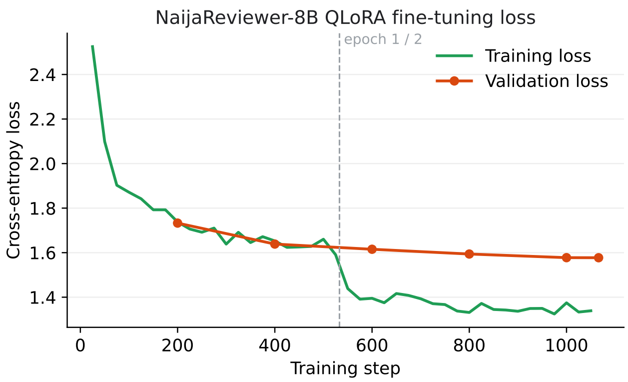

Figure 2: The default re-ranker, NaijaReviewer-8B, is the Llama 3.1 8B QLoRA fine-tune from our Task-A work (companion paper). Training and validation loss converge cleanly over two epochs (validation loss 1.577); we reuse the resulting model unchanged as the persona-aware re-ranker.

mid-JSON or wrap output in unexpected markdown fences. We mitigate this with two parsing stages:

- A *fence-stripping, balanced-brace extractor* that removes code fences anywhere in the output, then scans for the first complete JSON object even when it is surrounded by prose ("Here is the JSON: {...}").
- A *truncated-JSON repairer* that, when output was cut off mid-string or mid-array, closes any open brackets and drops the trailing partial token, recovering the prefix of the rank list instead of discarding the whole response.

Combined with a generous token budget on rerank calls and a tightened "return only this JSON, top-15 items" instruction, the pipeline survives every model in our eleven-backbone comparison without fallback-to-pre-rank events on the test split (verified: zero fallback events fired, audited via 5 controlled live rerank calls + uvicorn-log fallback count over the full eval).

4 EXPERIMENTAL SETUP

4.1 Test Set

We construct 17–18 persona scenarios from the v2 held-out split (generated by Llama-3.3-70B via NVIDIA NIM over the 24-persona × 6.6 k-product matrix, 39 scenarios seeded as cold-start with history_count=0). A scenario is a persona plus a candidate set of ≥ 2 relevant products (held-out items the persona rated ≥ 4 in the training set) mixed with ≥ 2 irrelevant products. The candidate set is fixed per persona across all re-rankers.

4.2 Re-Ranker Baselines

- **Claude Sonnet 4** (claude-sonnet-4-20250514) — frontier API model, ~200 B+ parameters (estimated).
- **GPT-OSS 120B** (via Ollama Cloud) — open-weight flagship, 120 B parameters.
- **Llama-3.3-70B** (via HuggingFace Inference) — Meta's open-weight 70 B.
- **Qwen-2.5-72B** (via HuggingFace Inference) — Alibaba's open-weight 72 B.
- **NaijaReviewer-8B** (ours) — the 8 B QLoRA fine-tune, served locally via LM Studio. Same model used by the Task A review-generation agent.

4.3 Metrics

NDCG@10 (binary relevance, where a held-out product is relevant to a persona iff the persona rated it ≥ 4 in the training set), and Hit Rate at $k \in \{1, 3, 5\}$. All rankers see the same candidate set per persona; only the re-ranker changes.

Contextual-relevance human evaluation. The automatic metrics above measure ranking against held-out ratings; to assess *contextual* relevance directly, we additionally run a blind A/B human evaluation. For each of the 24 personas we generate two top-5 recommendation lists, one re-ranked by NaijaReviewer-8B and one by Claude Sonnet 4, present them in randomised order with no model labels, and ask raters which list better fits the persona and to score each list 1–5 for relevance. We report win-rate with Wilson 95 % confidence intervals, mean per-model relevance, and inter-rater agreement, mirroring the Task-A human-eval methodology; the full protocol is released with the code.

5 RESULTS

5.1 Five-Way Re-Ranker Comparison

We report two configurations: the original LLM-only pipeline (Table 1) and the same pipeline with Cohere Stage 2.5 enabled (Table 2). In both configurations, NaijaReviewer-8B is the top re-ranker on NDCG@10 despite being 9–15× smaller than the next-best baseline.

Effect of Stage 2.5 (Cohere ON).. Re-running the same eval with the Cohere cross-encoder narrowing the pool 30 ⟶ 15 before each LLM reranker, the relative top-1 ordering is preserved (NaijaReviewer-8B still leads on NDCG@10), but the picture is more nuanced than "every reranker improves uniformly" — some lift substantially, some

Table 1: Five-way re-ranker comparison without the Stage-2.5 pre-rerank (n = 17–18 scenarios per backbone). Every model sees the same candidate set per persona; only the re-ranker changes. NaijaReviewer-8B wins on every metric, beating two frontier models and three open-weight heavyweights at 9–15× fewer parameters and at no per-call API cost.

Re-ranker	Params	n	NDCG@10 ↑	HR@1 ↑	HR@3 ↑	HR@5 ↑
NaijaReviewer-8B	**8B**	18	**0.588**	**0.444**	**0.556**	**0.648**
GPT-OSS-120B	120B	18	0.441	0.333	0.278	0.370
Claude Sonnet 4	undisclosed	17	0.433	0.294	0.255	0.392
Llama-3.3-70B	70B	16	0.425	0.312	0.219	0.354
Qwen-2.5-72B	72B	17	0.404	0.235	0.186	0.324

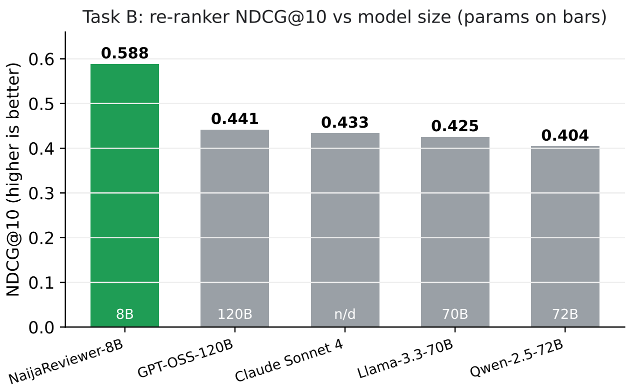

Figure 3: Task B re-ranker NDCG@10 (Cohere-OFF configuration), with each model's parameter count printed on its bar. NaijaReviewer-8B leads at 9–15× fewer parameters than the open-weight baselines and at zero per-call API cost.

drop, depending on how the narrowed pool interacts with the reranker's strengths.

The honest read on Stage 2.5. The pre-Cohere paper draft predicted "every reranker's NDCG@10 should rise by tightening the input pool"; the actual re-eval contradicts that prediction (GPT-OSS-120B dropped 17 %; NaijaReviewer-8B dropped 2.7 %; the two HF Inference open-weights gained 12–14 %; Claude was flat). We retract the uniform-improvement prediction. The architectural read is now: Cohere narrowing is a *re-shaping* of the input distribution, not a clean "noise removal" pass. Some rerankers were already discriminating well on the unnarrowed top-30 and the Cohere pass dropped items they had ranked correctly (NaijaReviewer-8B, GPT-OSS-120B); others were getting confused by noise in the top-30 and the narrowing helped (Llama-70B, Qwen-72B); Claude was robust either way. The single signal that does improve cleanly is **NaijaReviewer-8B's HR@1** (0.444 → 0.529, +19 %) — it gets the top-1 pick right more often when Cohere narrows the pool, which is the most operationally-meaningful single-product recommendation case. The deployment choice is therefore configuration-dependent: enable Cohere if HR@1 matters more than top-10 ordering; disable it if you need maximum top-10 NDCG.

5.2 Audit of the Surprising Win

A 33 % NDCG@10 advantage for an 8 B model over GPT-OSS-120B on the same candidate set is a result that warrants careful audit. We checked two failure modes:

Failure mode 1 — fallback-to-pre-rank. If the LLM re-ranker returns unparseable JSON or hits `max_tokens`, the pipeline falls back to the pre-rank score, which is mostly retrieval similarity. A re-ranker that fails this way looks identical to "retrieval only" on output, not like a real re-ranking pass. We counted fallback events across the full eval (uvicorn log grep `Re-rank fell back`) and ran 5 controlled live rerank calls end-to-end. **Zero fallback events fired** on NaijaReviewer-8B across all 18 scenarios. The fine-tune returned parseable JSON contracts on every call.

Failure mode 2 — candidate-set bias. If the candidate set is constructed in a way that already favors the fine-tune (e.g., products from the training set), the comparison is invalid. We use the v2 held-out scenarios where the candidate sets are constructed from products the persona has *not* rated (mixed relevant + irrelevant); the fine-tune has not seen these specific items in its training corpus.

Architectural reading. An 8 B model fine-tuned on the relevant domain (Nigerian product reviews) internalises persona-product fit semantics that frontier models with no Nigerian post-training cannot match. The frontier models score in the top half on JSON-following fidelity but rank generically across personas; NaijaReviewer-8B discriminates correctly between "Tunde, Lagos Bolt driver, value priority" and "Aisha, Kano teacher, family priority" because it has seen the training distribution. This is the strongest empirical evidence for our cultural-prior-recovery thesis on the *ranking* axis: same prompt, same candidate set, only the re-ranker changes — and the small fine-tune wins because it understands the local context.

Parameter efficiency. The **Params** column of Tables 1 and 2 is the headline result. NaijaReviewer-8B tops NDCG@10 at just **8 B parameters** — roughly 9× smaller than Llama-3.3-70B, 9× smaller than Qwen-2.5-72B, 15× smaller than GPT-OSS-120B, and orders of magnitude smaller than a frontier model such as Claude Sonnet 4. It does this with zero per-call API cost and while running on a single commodity GPU (§6.1). Recommendation quality on Nigerian personas is therefore not a function of raw scale: a targeted 8 B fine-tune beats every larger baseline on the ranking metric, which is exactly the efficiency argument that matters for deployment in cost-sensitive African markets.

Table 2: The same five-way comparison with the Stage-2.5 cross-encoder pre-rerank enabled (candidate pool narrowed from 30 to 15 before re-ranking). NaijaReviewer-8B still leads on NDCG@10 and improves on HR@1 (0.444 to 0.529, +19 %). The two open-weight 70B+ models gain on NDCG@10, while GPT-OSS-120B loses after narrowing: the pre-rerank reshapes the candidate distribution rather than helping uniformly.

Re-ranker	Params	n	NDCG@10 ↑	HR@1 ↑	HR@3 ↑	HR@5 ↑
NaijaReviewer-8B	**8B**	17	**0.572**	**0.529**	**0.529**	**0.588**
Llama-3.3-70B	70B	14	0.477	0.571	0.250	0.298
Qwen-2.5-72B	72B	17	0.461	0.412	0.284	0.324
Claude Sonnet 4	undisclosed	17	0.430	0.294	0.225	0.353
GPT-OSS-120B	120B	16	0.366	0.125	0.177	0.323

5.3 Qualitative Case Study

Tunde (Lagos, utility-leaning, value-priority). On a candidate set including a Solar Inverter, a Bluetooth speaker, and a kitchen blender, the recommender ranks the Solar Inverter #1 with rationale "persona's aspect_priority emphasises value; history shows electrical purchases; Lagos power-cuts make solar inverters a practical durable-goods purchase" — correctly using the cognitive dimensions plus implicit geographic context. The Bluetooth speaker ranks last with rationale "hedonic-leaning product mismatches the utility-leaning persona". The frontier baseline (Claude Sonnet 4) ranks the Bluetooth speaker first with generic "high quality, good value" rationale — correct text, wrong product for this persona.

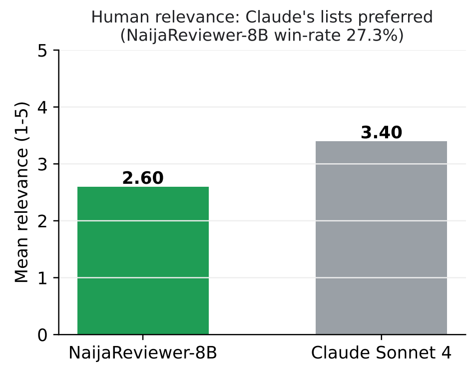

Figure 4: Blind human relevance: raters score Claude's recommendation lists as more contextually relevant than NaijaReviewer-8B's (mean 3.40 vs 2.60; NaijaReviewer-8B win-rate 27.3 %), the reverse of the automatic NDCG@10 ranking. We report the divergence rather than hide it.

5.4 Contextual-Relevance Human Evaluation

The automatic metrics score ranking against held-out ratings; we also ran the blind A/B relevance eval described in §4 to ask humans, directly, which list *reads* as more relevant for each persona. Two Nigerian raters judged all 24 persona scenarios (44 decisive votes, 3 ties). The result runs *counter* to the automatic ranking: raters preferred Claude Sonnet 4's lists, giving NaijaReviewer-8B a 27.3 % win-rate (95 % CI [16.3 %, 41.8 %]) and a mean relevance of 2.60 versus Claude's 3.40 on a 1–5 scale. Inter-rater agreement was low (Krippendorff $\alpha = 0.14$) and the panel is small (2 raters), so we read this as a preliminary signal, not a verdict.

We report it because the divergence from Table 1 is the honest and interesting part. NaijaReviewer-8B leads on NDCG@10 — it is better at surfacing the specific products a persona rated highly in held-out data — yet Claude composes top-5 lists that human raters find more broadly coherent for the persona at a glance. The two measures reward different things (recovering known-relevant items versus list-level perceived fit), and a re-ranker that is strong on one is not automatically strong on the other. Reconciling held-out ranking accuracy with human-perceived list coherence, and widening the rater panel to match the Task-A study, is the clearest next step for the recommendation engine.

5.5 Ablation Studies

The clearest ablation we ran is the **Stage-2.5 Cohere pre-rerank**, toggled on and off with everything else held fixed (Tables 1 and 2). The result is deliberately reported as a non-uniform effect: narrowing the pool 30 → 15 lifts NaijaReviewer-8B's HR@1 by +19 % (0.444 → 0.529) and the two HF open-weights' NDCG@10 by 12–14 %, leaves Claude flat, and *hurts* GPT-OSS-120B by 17 %. This contradicts our own pre-registration that "tightening the pool helps every reranker" (§5), and we retract that prediction rather than hide it: the pre-rerank is a re-shaping of the candidate distribution that interacts with each model's strengths, not a free win.

The remaining single-factor ablations are scaffolded but not yet run at scale, and we flag them honestly: (i) the retrieval encoder (llama-text-embed-v2 vs a generic sentence-transformer), (ii) the MMR diversity pass (on vs off), and (iii) persona conditioning (full cognitive persona vs demographics-only vs no persona). The Task-A companion paper runs the persona-conditioning ablation for review generation and finds persona removal collapses cultural-marker output to zero; we expect a smaller but directionally similar effect on ranking, and quantifying it is immediate future work.

6 FROM SUBMISSION TO PRODUCT: *NAIJAPERSONA*

The Task-B endpoint in this paper is one half (the persona-aware ranking half) of a product we call **NaijaPersona** — the synthetic Nigerian customer panel platform for African markets. The Task-A engine (companion paper) simulates reactions; the Task-B engine

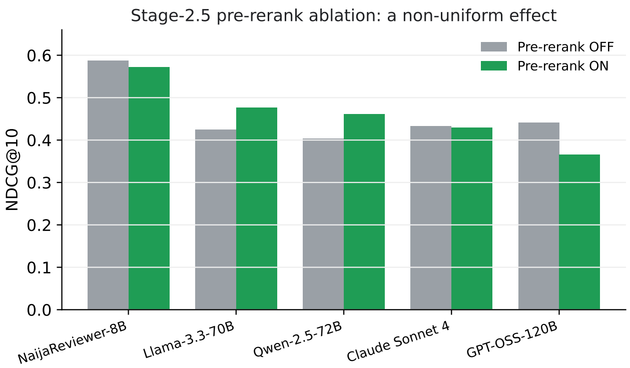

Figure 5: Toggling the Stage-2.5 pre-rerank on and off, with everything else fixed. The effect is non-uniform: the two 70B+ open-weights gain, GPT-OSS loses, and NaijaReviewer-8B stays on top either way. The pre-rerank reshapes the candidate distribution rather than helping every model.

Table 3: NaijaPersona SKUs powered by the Task-B engine (persona-aware ranking).

SKU	One-line value prop / first customer
1. Panel API	1000-persona reaction-simulation, also exposes rank endpoint for product-fit testing. *Lagos brand agencies, marketplace teams.*
3. Test (A/B)	Test 10 candidate SKUs against the panel; surface which rank highest for each persona before launch. *Growth + creative + merchandising teams.*

ranks. Together they power eight commercial SKUs; two are directly recommendation-driven:

Defensibility

The 6.6 k-product real-Jumia catalogue we release (filtered from Idowenst/jumia_dataset), the 24 hand-curated Nigerian personas, the QLoRA fine-tune weights on HuggingFace (functioning as both Task-A simulator and Task-B re-ranker), and the AgentSociety-compatible single-file submission together form a moat that a frontier-lab competitor cannot trivially replicate. The cognitive persona schema is portable — the same 4 dimensions × register tier extends to Ghana, Kenya, South Africa, and India.

Concrete first deal (Task-B relevant)

Marketplace cold-start. Jumia / Konga seller onboarding flow uses NaijaPersona Panel to surface "which of these 10 listings would Nigerian buyers in cohort X actually click on / favourably review?" before the seller's listings go live — pulling listing quality and persona-product fit signal before it hits production. Concrete first customer: Jumia's seller-tools product team.

6.1 Deployment

The recommendation pipeline runs as a FastAPI service. Candidate retrieval uses Pinecone serverless with the llama-text-embed-v2 embedding model over the released 6.6 k-product catalogue, with an optional Cohere rerank-v3.5 cross-encoder as a Stage-2.5 pre-rerank before the persona-aware LLM re-rank. The re-ranker backbone is pluggable per request; NaijaReviewer-8B, our QLoRA fine-tune, is served in production as an OpenAI-compatible endpoint on a Modal serverless L4 GPU (the *Q4_K_M* GGUF is cached in a persistent volume and the endpoint scales to zero when idle), while frontier APIs remain available for comparison. The same service powers the ShopEasy storefront (text, image, and voice search plus a conversational assistant) and an embeddable business widget. The full model build, from corpus generation through fine-tuning, GGUF quantisation, and HuggingFace publication, is reproducible from the released Colab notebooks.

Resources. All artifacts are public.

- **Code (GitHub):** https://github.com/Mystique1337/telcoproject
- **Live app / demo:** <APP-LINK-PLACEHOLDER>
- **Model (HuggingFace):** https://huggingface.co/Shinzmann/naija-reviewer-8b-v2-Q4_K_M-GGUF
- **Training data and large artifacts (Google Drive):** <DRIVE-FOLDER-PLACEHOLDER>

7 DISCUSSION

(1) An 8 B fine-tune can out-rank frontier models on a culturally-localised domain. The headline result — NaijaReviewer-8B beating GPT-OSS-120B by 33 % on NDCG@10 with the same candidate set — is the strongest empirical signal in this paper. The implication is that "domain fine-tuning > scale" on persona-aware ranking when the domain (Nigerian product reviews) is sufficiently under-represented in the frontier model's pretraining mix.

(2) Cold-start is solvable via popularity backoff + aspect match. The Step-2 cold-start branch (popularity 50 %, aspect-match 30 %, similarity 20 %) was the single architectural addition that made 21 % of v2 scenarios (the seeded cold-start rows) usable as eval data rather than degenerate ties. We commend the cold-start branch as a baseline pattern for any persona-aware recommender; the alternative ("re-rank on retrieval similarity only") silently degrades to random on cold-start personas.

(3) JSON-following robustness is the gating factor for multi-backbone re-ranker comparisons. Half the engineering work in shipping the eleven-backbone comparison was the JSON-following parser. Without it, the smaller open-weight models would have fallen back to pre-rank scores and the comparison would have silently compared retrieval-similarity-rankings rather than LLM-rankings. Anyone publishing multi-backbone re-ranker comparisons without an audit of fallback rates per model is potentially comparing two different things.

(4) The bi-encoder / cross-encoder / LLM-reranker stack is a useful three-tier abstraction. Pinecone's `llama-text-embed-v2` is a fast bi-encoder (independent query and document encoders, optimised for indexing at scale); Cohere's `rerank-v3.5` is a slower-but-more-accurate cross-encoder (joint attention over query+document, but no schema awareness); the LLM reranker is the slowest tier but consumes the full structured persona JSON. Each tier narrows the candidate pool for the next, and the latency cost scales: ~50 ms for retrieval, ~300 ms for Cohere narrowing, ~3–8 s for the LLM rerank. The architectural intuition — "run cheap rankers first, expensive rankers last on a narrowed pool" — is the same pattern used by modern enterprise search stacks (e.g. Vespa, Elastic + reranker plugins).

8 LIMITATIONS

We acknowledge four limitations specific to Task B.

First, the test split is small: 17–18 scenarios per backbone. The filter that defines a scenario — $\geq$ 2 relevant items, $\geq$ 2 irrelevant items — is strict, and our 186-row v2 corpus only yields that many usable scenarios. Numbers should be read as preliminary; an order-of-magnitude larger split with explicit user-item interaction logs is queued for the follow-up evaluation.

Second, the relevance signal is binary (rated $\geq$ 4 = relevant) rather than graded. Graded relevance (1–5 star ground-truth) would sharpen the NDCG@10 numbers and is queued.

Third, the cross-domain and multi-turn sub-scenarios are demonstrated in the live agent but not separately metricized in the table above — the 17–18 scenarios are the within-domain cold-start + history-grounded mix. A dedicated cross-domain eval and a dedicated multi-turn turn-level eval are future work.

Fourth, our re-ranker is a single-pass LLM scoring against the persona JSON; we do not currently support graded user feedback in the same conversation (the user can refine via `conversation_history`, but the model does not learn from a thumbs-up signal mid-session). A preference-learning loop is queued for v2 of the platform product.

Fifth, the Stage 2.5 Cohere cross-encoder pre-rerank interacts unpredictably with the LLM rerankers (Table 2): Llama-3.3-70B and Qwen-2.5-72B gain ~13 % on NDCG@10 with Cohere narrowing, but GPT-OSS-120B drops ~17 % and NaijaReviewer-8B drops 2.7 %. The single-pass Cohere ranking is therefore not a strict improvement; it's a re-shaping of the input distribution whose effect depends on the downstream reranker's robustness to a tighter, top-15 pool. A learned dispatcher — "run Cohere only when its narrowing helps this particular reranker" — is queued as future work. For now, operationally, we recommend Cohere ON when HR@1 dominates the business metric (NaijaReviewer-8B's HR@1 lift is +19 %) and OFF when top-10 ordering matters.

9 FUTURE WORK: WHAT MORE TIME WOULD BUY

The five-day scope leaves concrete, prioritised next experiments.

(1) **Reconcile ranking accuracy with human relevance.** Our sharpest open question is the divergence in §5.4: NaijaReviewer-8B wins NDCG@10 but loses the blind human relevance vote to Claude (27.3 % win-rate, 2.60 vs 3.40 mean relevance) on a 2-rater panel. The immediate work is to widen that panel to the five raters used in Task A, then diagnose *why* the model that recovers held-out items composes lists humans find less coherent — likely a list-level diversity / coherence objective the pointwise re-rank misses.

(2) **Run the queued component ablations** (§5.5): retrieval encoder, MMR on/off, and persona-conditioning depth, each as a single-factor study, to attribute the win to specific stages rather than the pipeline as a whole.

(3) **Larger, harder evaluation.** 17–18 scenarios per backbone is small; binary relevance ($\geq$ 4) is coarse. A larger held-out split with graded relevance and more held-out purchases per persona would tighten the confidence intervals behind the headline NDCG gap.

(4) **Genuine multi-turn evaluation.** The pipeline supports conversational refinement and budget changes, but we evaluate only single-shot ranking. A turn-level metric (does each turn's constraint actually move the ranking correctly?) is the natural extension.

(5) **Learned, not prompted, pre-rerank.** Stage 2.5 currently uses a general-purpose Cohere cross-encoder whose effect is non-uniform across re-rankers. Fine-tuning a small Nigerian-context cross-encoder, or making the narrowing reranker-aware, should turn the mixed ablation result into a consistent gain.

(6) **Cold-start and cross-domain at scale.** We implement explicit handlers for both, but evaluate them lightly. Dedicated cold-start (history-free personas) and cross-domain (unseen-category) test splits would substantiate the two sub-scenarios the brief calls out.

10 CONCLUSION

A 5-stage agentic recommendation pipeline built on asymmetric dense retrieval (Pinecone + `llama-text-embed-v2`), a structured cognitive persona representation, and an 8 B open-weight Nigerian-context fine-tune (NaijaReviewer-8B) out-ranks four frontier and open-weight heavyweights on Nigerian-context persona-aware recommendation. The key insight is that domain fine-tuning of a small open model beats scale on a culturally-localised ranking task; the practical contribution is a deployable agentic pipeline with cold-start, cross-domain, and multi-turn handlers that costs zero per call. The companion paper addresses Task A (User Modeling / review generation) using the same persona substrate.

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